

Case Study: The Cost of Cross-Framework Blindness in PARP Inhibitor Development

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1. Overview

The PARP inhibitor field is one of the most successful therapeutic programs in modern oncology. Poly(ADP-ribose) polymerase (PARP) inhibitors exploit synthetic lethality — the principle that cells with defective homologous recombination DNA repair (e.g., BRCA1/2 or RAD51C mutations) are selectively killed by PARP inhibition [Farmer et al. 2005, Bryant et al. 2005]. Six PARP inhibitors have received regulatory approval. Thousands of patients have benefited. The field’s contribution to cancer treatment is not in question.

What is in question is the citation infrastructure that supported the field’s development. We present three layers of documented citation failure, each demonstrating a different component of the confidence chain problem, each verifiable from publicly available data, and each addressable by the confidence chain method proposed in the companion paper.

The purpose of this case study is not to diminish any individual contribution. It is to demonstrate, on one of the most impactful therapeutic programs of the century, that the current system for tracking scientific provenance leaves knowledge on the table — with costs measured not in citation counts but in years of delayed treatment and the patients who did not have those years.

2. Layer 1: Prior Art Not Cited — A Twenty-Five Year Conceptual Foundation

The landmark papers

Two papers published simultaneously in *Nature* in April 2005 established PARP inhibition as a therapeutic strategy for BRCA-deficient cancers:

- Farmer H, McCabe N, Lord CJ, Tutt AN, Johnson DA, Richardson TB, Santarosa M, Dillon KJ, Hickson I, Knights C, Martin NM, Jackson SP, Smith GC, **Ashworth A**. “Targeting the DNA repair defect in BRCA mutant cells as a therapeutic strategy.” *Nature*. 2005;434:917-921. PMID: 15829967. **Cited by 3,298 papers** (as of April 2026).

- Bryant HE, Schultz N, Thomas HD, Parker KM, Flower D, Lopez E, Kyle S, Meuth M, Curtin NJ, **Helleday T**. “Specific killing of BRCA2-deficient tumours with inhibitors of poly(ADP-ribose) polymerase.” *Nature*. 2005;434:913-917. PMID: 15829966. **Cited by 2,633 papers.**

Combined citations: **5,931**. These papers are universally cited as the origin of the synthetic lethality approach to cancer therapy using PARP inhibitors. Bryant et al. described the finding as representing “a new concept in cancer treatment.”

The prior art — twenty-five years of conceptual foundation

The conceptual framework within which Farmer and Bryant’s experiment operates was assembled across multiple fields over twenty-five years. None of the following prior contributions are cited in the landmark 2005 Nature papers for the conceptual claims they establish.

PARP inhibitor sensitization (1980–2004). The first PARP inhibitor, 3-aminobenzamide, was described by Purnell and Whish in 1980. Throughout the 1980s and 1990s, PARP inhibitors were used as standard tools for studying DNA repair, with their ability to sensitize cells to DNA-damaging agents well documented. By 2004, **29 papers** in PubMed described PARP inhibitor sensitization in the context of DNA repair and cancer, including:

- Veuger SJ, Curtin NJ, Richardson CJ et al. “Radiosensitization and DNA repair inhibition by the combined use of novel inhibitors of DNA-dependent protein kinase and poly(ADP-ribose) polymerase-1.” *Cancer Res.* 2003;63(18):6008-6015. PMID: 14522929.
- Calabrese CR, Almassy R, Barton S et al. “Anticancer chemosensitization and radiosensitization by the novel poly(ADP-ribose) polymerase-1 inhibitor AG14361.” *J Natl Cancer Inst.* 2004;96(1):56-67. PMID: 14709739.
- Curtin NJ, Wang LZ, Yiakouvaki A et al. “Novel poly(ADP-ribose) polymerase-1 inhibitor, AG14361, restores sensitivity to temozolomide in mismatch repair-deficient cells.” *Clin Cancer Res.* 2004;10(3):881-889. PMID: 14871963.

The concept that PARP inhibition sensitizes cancer cells to DNA damage was not new in 2005. It was two decades old and documented in dozens of papers.

PARP biology and DNA repair (1992–2004). The de Murcia laboratory in Strasbourg characterized PARP-1’s structure and function in DNA repair beginning in 1992:

- Schreiber V, Molinete M, Boeuf H, de Murcia G, Ménissier-de Murcia J. “The human poly(ADP-ribose) polymerase nuclear localization signal is a bipartite element functionally separate from DNA binding and catalytic activity.” *EMBO J.* 1992;11(9):3263-3269. PMID: 1505517.
- Masson M, Niedergang C, Schreiber V, Muller S, Menissier-de Murcia J, de Murcia G. “XRCC1 is specifically associated with poly(ADP-ribose) polymerase and negatively regulates its activity following DNA damage.” *Mol Cell Biol.* 1998;18(6):3563-3571. PMID: 9584196.

The XRCC1-PARP interaction in base excision repair — the mechanistic foundation for understanding why PARP inhibition causes DNA damage accumulation — was established by 1998, seven years before the Nature papers.

PARP and BRCA in the same repair pathway (2000). The biological connection between PARP and BRCA was published five years before Farmer/Bryant:

- Lansdorp PM. “Repair of telomeric DNA prior to replicative senescence.” *Mech Ageing Dev.* 2000;118(1-2):23-34. PMID: 10989122.

This paper lists ATM, p53, PARP, DNA-PK, Ku70/80, the hRad50-hMre11-p95 complex, **and BRCA 1 and 2** in the same sentence as molecules involved in the repair of telomeric DNA. The two molecules — PARP and BRCA — were already understood to operate in the same repair space in 2000.

Synthetic lethality as a cancer therapeutic strategy (2002–2003). The concept of exploiting synthetic lethal interactions for cancer treatment was proposed explicitly:

- **Garber K.** “Synthetic lethality: killing cancer with cancer.” *J Natl Cancer Inst.* 2002;94(22):1666-1668. PMID: 12441317. **Cited by 8 papers.**

A news article in JNCI reporting that researchers were already exploring synthetic lethal interactions as a therapeutic strategy for cancer. The concept was visible enough to a science journalist in 2002 that it warranted coverage in the National Cancer Institute’s own journal.

- **Kamb A.** “Mutation load, functional overlap, and synthetic lethality in the evolution and treatment of cancer.” *J Theor Biol.* 2003;223(2):205-213. PMID: 12814603. **Cited by 8 papers.**

A formal theoretical paper proposing that synthetic lethal interactions between mutated genes in cancer cells could be therapeutically exploited. Kamb specifically argued that “drugs that target complex processes that utilize genetically redundant or overlapping components, such as DNA replication and chromosome segregation, offer attractive target opportunities” — the same biological territory (DNA replication and repair) that Farmer and Bryant would demonstrate experimentally two years later.

- **Garber K.** “Running interference: pace picks up on synthetic lethality research.” *J Natl Cancer Inst.* 2004;96(13):982-983. PMID: 15240774.

A follow-up news article in JNCI, one year before the Nature papers, reporting accelerating research activity in synthetic lethality for cancer.

The concept of synthetic lethality itself has a long history in genetics, dating to Dobzhansky’s work in *Drosophila* (1946) and Lucchesi’s formalization (1968). By 2004, **225 papers** containing the phrase “synthetic lethality” or “synthetic lethal” had been indexed in PubMed. Three of these specifically addressed cancer therapeutic applications. None were cited by the landmark 2005 Nature papers.

PARP inhibition sensitizing breast cancer cells (2005, simultaneous with Farmer/Bryant).

In the same year as the Nature papers, the de Murcia laboratory — which had been doing PARP biology for thirteen years — published:

- Muñoz-Gómez JA, Martín-Oliva D, Aguilar-Quesada R et al. “PARP inhibition sensitizes p53-deficient breast cancer cells to doxorubicin-induced apoptosis.” *Biochem J.* 2005;386(Pt 1):119-125. PMID: 15456408.

The same finding — PARP inhibition sensitizing breast cancer cells — published simultaneously by the lab that had established the foundational PARP biology. Farmer and Bryant

received Nature. The de Murcia lab received the Biochemical Journal. The field remembers Nature.

The citation gap

Neither Farmer et al. 2005 nor Bryant et al. 2005 cited Garber 2002, Kamb 2003, Garber 2004, or Lansdorp 2000. This is verifiable via PubMed citation data (NCBI E-utilities `pubmed_pubmed_citedin` link type, queried April 2026).

Decomposition Theorem classification

Under the classification system proposed in the companion paper, the full decomposition of Farmer and Bryant 2005 is:

Claim in the landmark papers	Classification	Prior art
PARP is involved in DNA repair	COLLAPSE	Established 1980s–1990s; de Murcia lab 1992+
PARP inhibitors sensitize cells to DNA damage	COLLAPSE	29 papers before 2005 (Curtin, Calabrese, Veuger et al.)
BRCA1/2 are involved in homologous recombination repair	COLLAPSE	Established 1990s
PARP and BRCA operate in the same repair space	COLLAPSE	Lansdorp 2000 (PMID 10989122)
Synthetic lethality can be exploited as cancer therapy	COLLAPSE	Kamb 2003; Garber 2002; 225 papers on synthetic lethality by 2004
PARP inhibition sensitizes breast cancer cells	COLLAPSE	de Murcia lab 2005 (PMID 15456408), simultaneous
Specific demonstration: PARP inhibitor on BRCA1/2-null cells with in vitro and in vivo survival data showing selective killing	NOVEL	No prior paper had demonstrated this specific drug-target-genotype combination
<i>“A new concept in cancer treatment”</i> (Bryant et al.)	Documentably false	The concept was not new at any level; the experiment was new

The current citation record does not distinguish between these classifications. Both the conceptual framework (which was assembled across multiple fields over twenty-five years) and the experimental

demonstration (which was genuinely novel) are attributed to Farmer and Bryant, with 5,931 combined citations. The prior conceptual work by Kamb has 8 citations. The PARP biology foundation by the de Murcia lab is credited for the biology but not for the therapeutic concept it enabled. The 29 papers on PARP inhibitor sensitization before 2005 are not credited for the principle Farmer and Bryant’s experiment validated. The distinction between the concept (which was not new) and the experiment (which was) is invisible in the published record.

What the confidence chain method would have caught

R1 (equivalency class declaration) would have required Farmer and Bryant to declare their base assumptions. The equivalency classes include: “PARP is involved in base excision repair” (EC with de Murcia 1992+), “PARP inhibition sensitizes cells to DNA damage” (EC with 29 prior papers), “BRCA is involved in homologous recombination” (EC with 1990s literature), “synthetic lethality as cancer therapeutic strategy” (EC with Kamb 2003, Garber 2002, the yeast genetics literature back to Dobzhansky 1946).

R2 (collapse/novel classification) would have required explicit classification: the conceptual contributions collapse into the prior literature; the specific PARP-BRCA experimental demonstration is novel.

R4 (provenance triples) would have traced each conceptual claim to its prior instances and made the distinction visible at the time of publication. The claim “a new concept in cancer treatment” would have been flagged as inconsistent with the provenance record.

3. Layer 2: Prior Art Noticed but Not Acted On

The cross-citing review papers

At least two review papers placed the prior conceptual work and the landmark experimental papers in the same reference list:

- **Reinhardt HC, Jiang H, Hemann MT, Yaffe MB.** “Exploiting synthetic lethal interactions for targeted cancer therapy.” *Cell Cycle*. 2009;8(19):3112-3119. PMID: 19755856. (MIT Koch Institute for Integrative Cancer Research.)

This review cites Kamb 2003 (PMID 12814603), Farmer 2005 (PMID 15829967), AND Bryant 2005 (PMID 15829966) in the same reference list. The authors at MIT had access to both the prior conceptual work and the landmark experimental papers. **They did not note that Farmer and Bryant had not cited Kamb.**

- **Barchiesi G, Roberto M, Verrico M, Vici P, Tomao S, Tomao F.** “Emerging Role of PARP Inhibitors in Metastatic Triple Negative Breast Cancer.” *Frontiers in Oncology*. 2021;11:769280. PMID: 34900718. (Sapienza University of Rome.)

This review cites both Kamb 2003 and Farmer 2005. Again, both in the same reference list. **Again, no note that the prior art was uncited by the landmark paper.**

What this demonstrates

The prior art gap was not invisible to the community. It was noticed — at least in the sense that review authors read both the prior work and the landmark papers and placed them in the same

reference list. **But the current system provides no mechanism for converting the noticing into a correction.** There is no structured way for a review paper to flag that a landmark paper’s conceptual contribution collapses into prior work. There is no process for updating the citation record of a published paper to reflect discovered prior art. There is no requirement for reviewers to check whether a claimed conceptual contribution has precedent in adjacent literature.

The noticing happened. The system had nowhere to put the noticing. The gap persisted for twenty-four years — not because it was hidden, but because the system has no infrastructure for converting noticed gaps into corrections.

The closest existing mechanism: meta-analysis

The nearest analog in the current system to cross-framework synthesis is the meta-analysis. Meta-analyses are retrospective — they synthesize what has already been published, sometimes years after the original publications. They do not correct the record: a meta-analysis that notices a prior art gap does not update the original paper’s citation record or propagate the finding to citing papers. Most critically, meta-analyses are not forward-looking — they do not flag what future papers should check. And in practice, forward-moving researchers do not consult meta-analyses when evaluating new claims; they cite the landmarks directly, inheriting whatever gaps the landmarks contain.

Meta-analysis is a patch applied after the fact that the community rarely integrates into its forward-looking evaluation of new contributions. The confidence chain method differs in that the synthesis is built into the infrastructure rather than performed as an occasional retrospective effort. The knowledge graph propagates corrections automatically. The equivalency class structure makes cross-framework checking tractable at review time.

What the confidence chain method would have caught

R2 (collapse/novel classification), if required of review papers, would have required Reinhardt 2009 to explicitly classify the relationship between Kamb 2003 and Farmer 2005 — to state whether the conceptual contribution was independent (both novel), overlapping (one collapses into the other), or contradictory. The classification would have been on the record in 2009, four years after the Nature papers and sixteen years before this analysis.

4. Layer 3: Cross-Field Knowledge Left on the Table

The two fields

Between 1992 and 1998, a research group at Vanderbilt University and a biotech startup called Advanced Therapies, Inc. (ATI) in Novato, California, developed cationic liposome- based gene delivery systems with synthetic intracellular localization peptides for targeted nuclear delivery of therapeutic payloads. Key outputs:

- A patent (WO1995034647A1, Conary and Brigham, Vanderbilt University, filed 1995): synthetic intracellular localization peptides for enhanced delivery of nucleic acids to the cell nucleus and mitochondria. **36 citing patents through 2017.**
- Eight peer-reviewed publications (Conary, Brigham, Schreier et al., 1993-1998) demonstrating aerosol and intravenous cationic liposome gene delivery to lungs, with applications to inflammatory disease, transplant rejection, and alpha-1-antitrypsin deficiency.

- An SBIR Phase I award (HHS, 1998, \$99,445) to Advanced Therapies, Inc. for “Synthetic Cell Delivery Systems for Polynucleic Acids.”

In parallel, the Strasbourg laboratory of Gilbert de Murcia characterized the nuclear localization signal (NLS) of PARP-1 itself — the same class of molecular mechanism the ATI patent was synthetically reproducing for therapeutic delivery:

- Schreiber V, Molinete M, Boeuf H, de Murcia G, Ménissier-de Murcia J. “The human poly(ADP-ribose) polymerase nuclear localization signal is a bipartite element functionally separate from DNA binding and catalytic activity.” *EMBO Journal*. 1992;11(9):3263-3269. PMID: 1505517.

The de Murcia laboratory later published directly on the ATM-PARP interaction and PARP inhibitor sensitization (Aguilar-Quesada et al. 2007, PMID 17459151), entering the lineage that would lead through the synthetic lethality discovery to the clinical PARP inhibitor program.

The PARP inhibitor clinical program

Beginning in 2009, Timothy Yap and colleagues at the Institute of Cancer Research / Royal Marsden and later MD Anderson Cancer Center conducted the first-in-human clinical trials of PARP inhibitors:

- Fong PC, Boss DS, **Yap TA**, Tutt A, Wu P, Mergui-Roelvink M, Mortimer P, Swaisland H, Lau A, O’Connor MJ, **Ashworth A**, Carmichael J, Kaye SB, Schellens JH, **de Bono JS**. “Inhibition of poly(ADP-ribose) polymerase in tumors from BRCA mutation carriers.” *New England Journal of Medicine*. 2009;361(2):123-134. PMID: 19553641.

Yap has published 56 papers on PARP inhibitors, of which 16 cite Farmer 2005. The saruparib (AZD5305) PARP1-selective inhibitor trial at MD Anderson treated patients with DNA repair gene mutations including RAD51C — a gene in the same homologous recombination repair pathway that the Vanderbilt/ATI group’s work was tangentially connected to through the shared nuclear localization signal mechanism.

The zero-overlap finding

We performed a systematic citation overlap analysis using NCBI E-utilities (April 2026). The combined forward citation footprint of all published Conary and Schreier papers (122 unique citing papers) was compared against the forward citation footprint of Farmer 2005 (3,298 citing papers).

The overlap is zero. Not one paper in the published scientific literature cites both a Conary/Schreier paper and Farmer 2005. The two citation communities are completely disjoint. The NLS/liposome delivery field and the PARP inhibitor field operated in different journals, different subfields, and different research communities for over thirty years without a single citation crossing.

The convergence that eventually happened

In 2019, twenty-four years after the ATI patent was filed, the two fields’ technologies finally merged:

- Zhang D, Baldwin P, Leal AS, Carapellucci S, Sridhar S, Liby KT. “A nano-liposome formulation of the PARP inhibitor Talazoparib enhances treatment efficacy and modulates immune

cell populations in mammary tumors of BRCA-deficient mice.” *Theranostics*. 2019;9(21):6224-6238. PMID: 31534547.

This paper places a PARP inhibitor (talazoparib) inside a liposomal delivery vehicle and tests it in BRCA-deficient breast tumors — combining the delivery technology the ATI group had pioneered with the therapeutic target the PARP inhibitor field had identified. The convergence was structural: both fields were working on the same problem (targeting molecular vulnerabilities in cancer cells via nuclear-directed interventions) from opposite ends, and the convergence arrived twenty-four years after the delivery technology was available because the two fields never read each other’s work.

What the delay cost

The liposomal PARP inhibitor approach, had it been attempted in 2005 rather than 2019, would have entered the clinical pipeline a decade earlier. Better formulation, better targeting, better bioavailability of PARP inhibitors may have improved outcomes for patients treated during the intervening years. The exact magnitude of the benefit is not quantifiable from the current data, but the structural observation is clear: knowledge that was available in one field was not used by another field for twenty-four years because the fields did not read each other’s literature, and the current system provides no mechanism for bridging such gaps.

What the confidence chain method would have caught

R1 (equivalency class declaration) would have required PARP inhibitor papers to declare their base assumptions about nuclear targeting and drug delivery. The NLS literature — both the PARP-1 NLS characterization (Schreiber 1992) and the synthetic NLS delivery work (Conary/Brigham patent, Schreier publications) — would have been identified as sharing an equivalency class with the implicit nuclear delivery assumptions of the PARP inhibitor field.

R5 (knowledge graph submission) would have made the cross-field connection queryable. A researcher preparing a PARP inhibitor paper could have queried the graph for “nuclear delivery of therapeutic payloads” and discovered the ATI/Vanderbilt delivery work without needing to know the specific authors, journals, or search terms.

5. Summary: Three Layers, One Problem

Layer	What happened	When it was discoverable	Current mechanism for correction	Confidence chain mechanism
1. Prior art not cited	Farmer/Bryant 2005 did not cite Kamb 2003 or Garber 2002 for the synthetic lethality cancer concept	At time of submission (a PubMed search)	None at review; letter to editor after publication (culturally discouraged, non-propagating)	R1 + R2: EC declaration and collapse classification at review time

Layer	What happened	When it was discoverable	Current mechanism for correction	Confidence chain mechanism
2. Prior art noticed, not acted on	Reinhardt 2009 and Barchiesi 2021 cited both Kamb and Farmer without noting the gap	2009 (four years after publication)	None; review papers have no mechanism for flagging prior art gaps in the papers they cite	R2 applied to review papers: explicit classification of relationships between cited works
3. Cross-field knowledge not bridged	Zero citation overlap between the NLS/delivery field and the PARP inhibitor field across 3,420 papers	At any point from 2005 onward (the NLS literature was indexed and searchable)	Meta-analysis (retrospective, non-propagating, rarely consulted by forward-moving researchers)	R1 + R5: EC declaration across fields + queryable knowledge graph

The cost

The cost of these three layers of citation failure is not measured in citation counts. It is measured in the delay between knowledge being available and knowledge being used. In Layer 1, a concept proposed in 2002-2003 was treated as novel in 2005 and has carried inflated novelty attribution for twenty-one years. In Layer 3, a delivery technology available since 1995 was not applied to PARP inhibitors until 2019 — a twenty-four year delay.

The patients who were treated during those years received the best available care. But “best available” was constrained by the field’s failure to compose across its own literature and across adjacent fields. The confidence chain method exists to close these gaps — not retroactively, but prospectively, by building the cross-framework synthesis into the infrastructure so that the next gap is shorter, and the next one shorter still, and eventually the gaps that cost patients years are measured in months instead of decades.

6. Methodological Note

All citation data in this case study was obtained from NCBI PubMed E-utilities (elink.fcgi with `pubmed_pubmed_citedin` and `pubmed_pubmed_refs` link types) on April 13, 2026. Citation counts are as of that date and may differ from counts reported by other sources (Google Scholar, Semantic Scholar) due to differences in indexing scope.

The zero-overlap finding in Layer 3 was computed by comparing the full forward citation sets of all published Conary JT papers (8 papers, 71 unique citers) and Schreier H delivery papers (10 papers, 70 unique citers; combined ATI footprint 122 unique papers) against the full forward citation set

of Farmer 2005 (3,298 citers). The comparison used sorted set intersection (UNIX `comm -12`). The analysis is fully reproducible using the PMIDs and methods described above.

The cross-citation analysis in Layer 2 was performed by checking the reference lists of Reinhardt 2009 (PMID 19755856) and Barchiesi 2021 (PMID 34900718) using NCBI E-utilities `pubmed_pubmed_refs` link type, confirming that both papers cite Kamb 2003 (PMID 12814603) and Farmer 2005 (PMID 15829967).

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